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Software Verification and Validation (DES3103)

Course Learning Outline

1. Briefing Session (Ice Breaking via Whiteboard, Class rules & Pro forma): **M. Teams**
2. Introduction to Software Verification & Validation I & II (V&V)
3. Managing V&V with SQA Integration
4. Quality metrics or attributes in V&V

WEEK 6-MID-TERM TEST (QUIZ): Topic 1 – 3-(CLO1:C2) – 20 MARKS

WEEK 7-PHASE 1: ASSIGNMENT 1-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO3:A5) - 20 MARKS: M. Teams

WEEK 9-PHASE 1: PRESENTATION & REFLECTION ON PHASE 1: M. Teams

5. Software Reviews – Will be covered on Assignments 1 & 2
6. Quality Control: Software Testing - Will be covered in Assignments 1 & 2
7. Human-Computer Interface Testing - Will be covered in Assignments 1 & 2
8. Manual Testing and Automation Testing: Will be covered in Final Assessment

WEEK 11-PHASE 2: ASSIGNMENT 2-ONLINE PROJECT-BASED LEARNING (OPBL)-(CLO2&4:P4&C5) - 20 MARKS: M. Teams
WEEK 13-PHASE 2: PRESENTATION & REFLECTION ON PHASE 2: M. Teams

9. Quality Maturity Models
10. Software Problem Analysis and Reporting
11. Quality Culture in doing V&V.
12. Safety and Security in V&V

WEEK 14: REFLECTION ON IMPLEMENTATION OPBL (SOFTWARE TESTERS) IN THE V&V COURSE: e-FORUM

WEEK 16-PHASE 3: FINAL ASSESSMENT – INDIVIDUAL SPECIAL PROJECT BASED ON PHASE 2 (CLO4:C5) (2 WEEKS) - 40 MARKS : e-ASSESSMENT

TOPIC 2

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Introduction to Software Verification & Validation (V&V)

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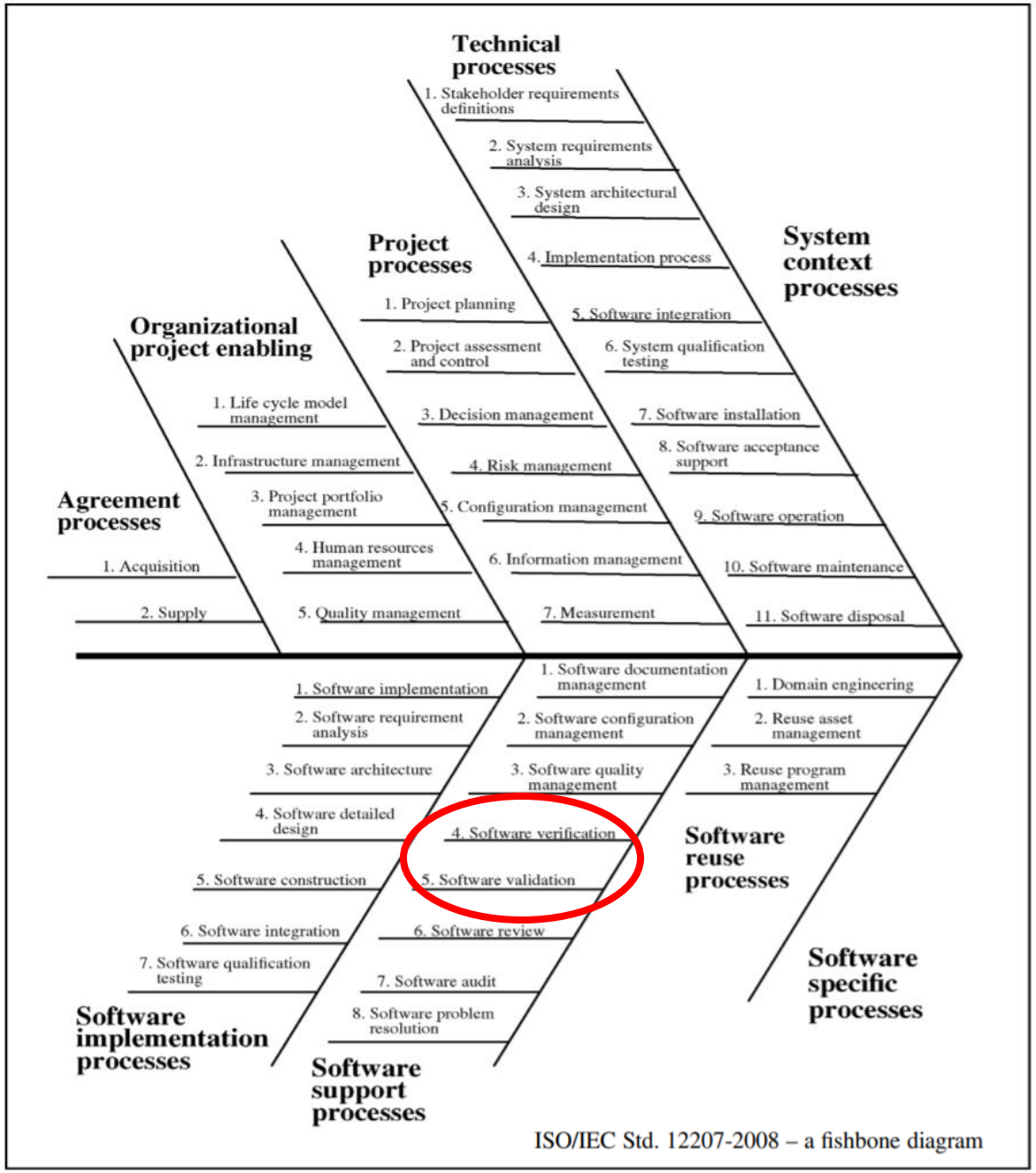
Introduction to Software Verification & Validation (V&V)

- The importance of conducting proper software V&V
  - Why is the V&V process important?
  - The rationale of having the V&V Process
  - Correctness / soundness / completeness



ISO/IEC Std. 12207-2008 – a fishbone diagram

Learning outcomes:

- Explain the importance of software verification & validation
- Work collaboratively in a team

Why is the V&V process important?

Why? V&V is needed because:

Software bugs could be expensive or even dangerous.

→ V&V is necessary because the designers and implementers of computer-based systems are human; they will make errors.

→ These errors will result in undetected faults in delivered systems.

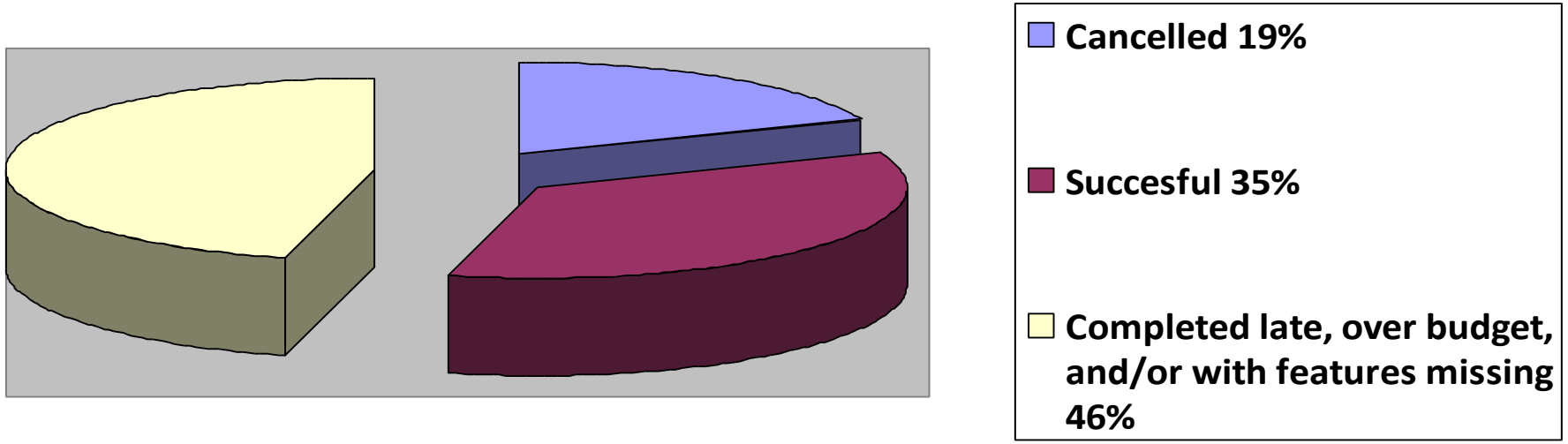
→ Faults can result in dangerous failures causing loss of life, financial loss **or** property damage.

→ The **mission** of V&V is to **find and correct errors as early as possible** in the development life cycle thus preventing the delivery of a faulty product to a customer.

- There were also many outcomes from software development projects reported with **features missing** or **functionality did not measure up to the claims** of the software developers.

Introduction to Software Verification & Validation (V&V)

The importance of conducting proper software (V&V)



A study by Standish Group, a research firm that analyses software development projects, have showed that the outcomes of over 9,000 development projects completed in 2006 were either cancelled, successful or completed late, over budget and/ or with features missing (Rubenstein, 2007).

In a survey conducted by the Cutter Consortium (2002), cases of functionality or performance of the software products delivered did not measure up to the claims of the software developers occurred as high as 67% from 78% of litigated IT companies. Too little software is delivered on time, within budget, fault free, and meeting its clients' needs.

The rationale of having the V&V process

Software professionals are human and sometimes make mistakes while developing a product. As a result there will be a fault in the software.

If a mistake is made while eliciting the requirements, the resulting fault will probably also appear in the specifications, the design, and the code. Clearly, the earlier we correct a fault the better. We must find faults early or else it will cost us money. Therefore, proper techniques should be employed to detect faults as early as during the requirements and analysis phases.

Early in the development life cycle, the product essentially exists only on paper. Hence, correcting faults during these phases may simply mean making a change to a document. It is far too late to check a software product once it is ready to be delivered to the client.

It is recommended that the process is carried out almost to the total exclusion of other activities. Usually, it occurs toward the end of each phase **(verification)** and will always happen before the product is handed over to the client **(validation).** If testing is treated separately **(testing phase),** it will not be carried out constantly throughout every product development and maintenance phase. This is incompatible with the goal of ensuring that a software product is as fault-free as possible at all times. There should never be times when no testing is being performed.

Correctness / Soundness / Completeness

- Part of the criteria for quality assessment (V&V).
- **Correctness -** accurately fulfills predefined criteria.
  - E.g.: The data attribute follows the requirements, the algorithm is correct with respect to a specification, each input produces the expected output, etc. (ERROR FREE).
- **Soundness -** In bug detection systems, soundness means the ability to detect all possible errors of a certain class. Preservation of program semantics = compilers
- **Completeness -** Requirements, Structure (code), and Architecture (data path/control link) are three software abstractions that are used to develop and build the software. These altogether can be described as the "size of the software" or COMPLETE.

Error, Fault and Failures

Software Error – made by programmer

- Syntax (grammatical) error
- Logic error (multiply instead of adding two operands)

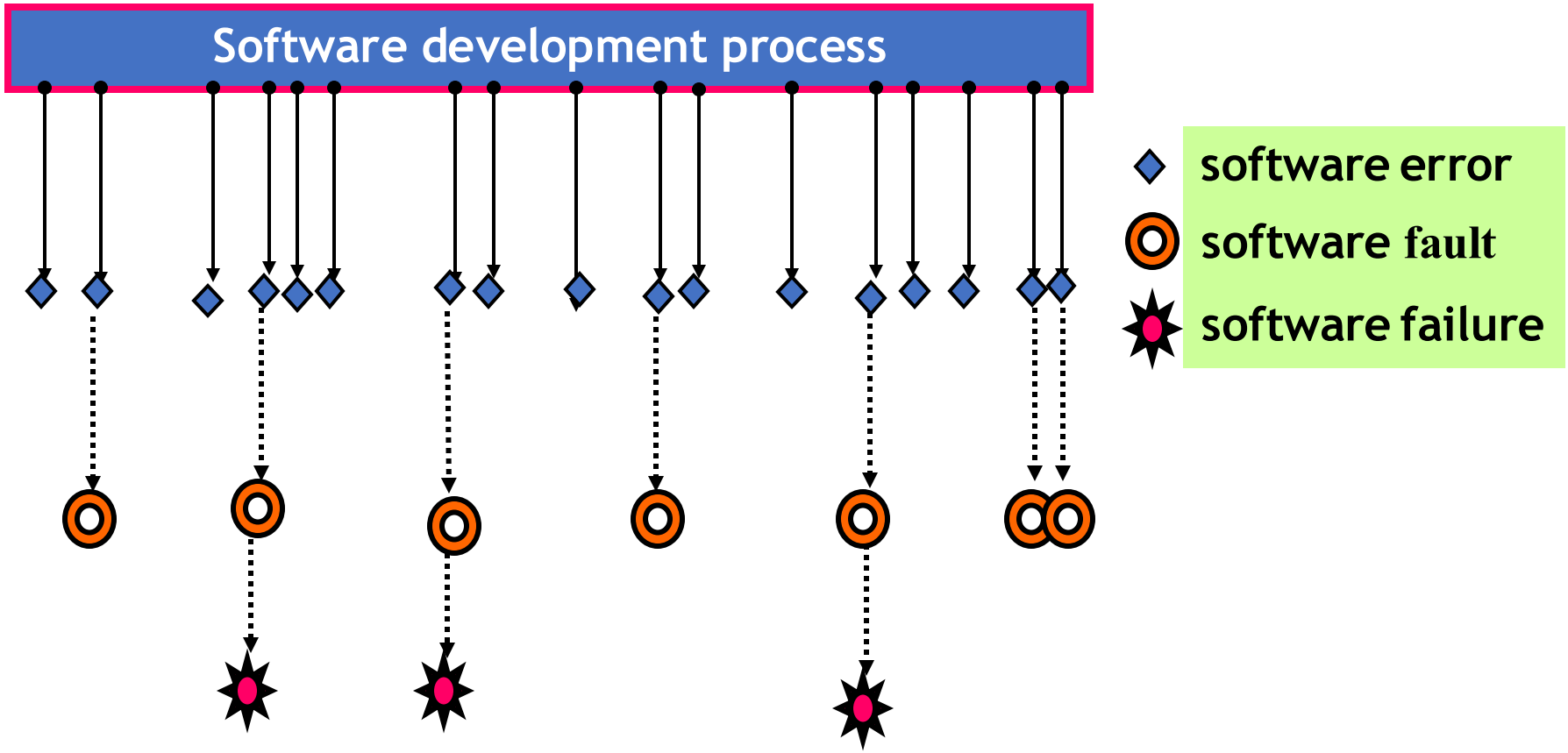
Software Fault

- All software errors may not cause software faults
- That part of the software may not be executed

Software Failures – Here's the interest.

- A software fault becomes a software failure when/if it is activated.
- Faults may be found in the software due to the way the software is executed or
- Other constraints on the software's execution, such as execution options.

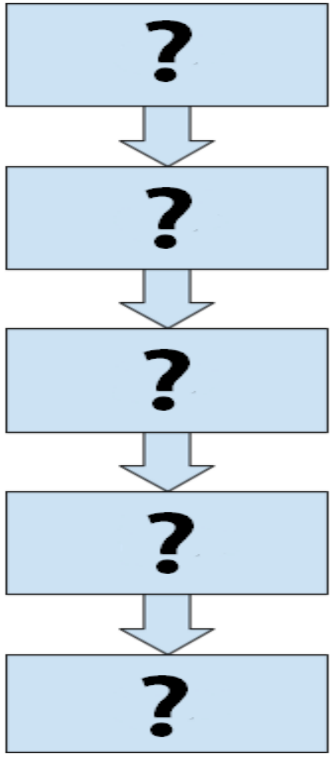
Error, Fault and Failures



Question:

Mistakes that need to find / correct / avoid in software development.
Match the elements according to ascending order (the most important).

Elements
Fault
Defects
Bug
Error
Failure



Introduction to Software Verification & Validation (V&V)

The importance of conducting proper software (V&V)

❖ **Fault**

- Faults are the result of an error. It is the incorrect step or process due to which the program or the software behaves in an unintended manner.
- If the fault is not activated, no error is produced.

❖ **Defect**

- A defect is said to be detected when Failure occurs.

❖ **Bug**

- A bug is evidence of a Fault in a program due to which the program does not behave in an intended manner.

❖ **Error**

- Error is a deviation from the actual and the expected result. It represents the mistakes made by the people. It refers to the human error that produces the problem due to incorrect thinking by the developer / programmer.

❖ **Failure**

- Failure is an inability of the system or components to perform its required function. Failure occurs when Faults execute.

Introduction to Software Verification & Validation (V&V)

Group Discussion : Differences between the Software Verification & Validation in terms of:

- Definition
- Evaluation period
- Objective
- Activities included
- Methods
- Items evaluated
- Others

References

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Thank You…
Terima kasih..